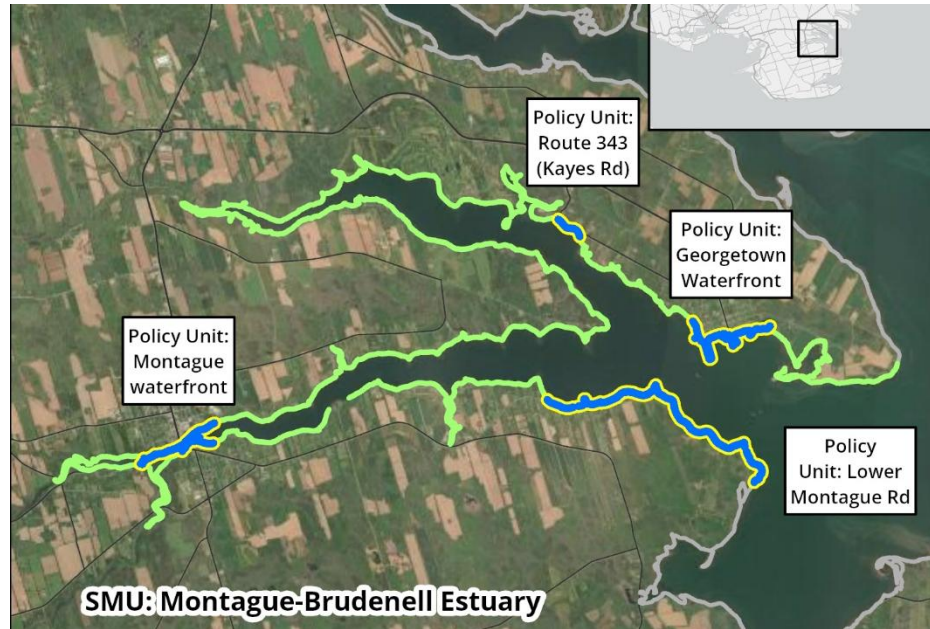


12. Darnley Basin

Shoreline length: 72 km



Shore Types

This SMU contains two estuaries (Montague to the south, Brudenell to the north). Both have crenulated shorelines with seaward headlands and protruding beach features. The main shore types are low plains and sloped armour, followed by wetlands.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
53	10	18	2	14	2	0	1

Governance, Infrastructure, and Economy

The Darnley Basin

SMU falls entirely within the Town of Three Rivers and includes both Georgetown and Montague.

The area includes coastal roads Dewars Road, Kent Street, Lower Montague Road, Riverside Drive, Roma Point Road, St Andrews Point Road, Station Street, Thornton Road, Valleyfield Road (Route 326), and Route 4/ Water Street.

Coastal infrastructure includes the Georgetown wharf, marina, pier, Georgetown West Street Beach, Montague Waterfront Marina, the Montague fishing wharf, and the Lower Montague wharf.

Coastal economic drivers would include the Georgetown and Montague waterfronts generally, Seafood 2000 Ltd fish plant, the Brudenell Canoe Kayak Club, Brudenell River Golf Course, the Rodd Brudenell Resort, Montague Cold Storage, and Beck Home Hardware.

Local and legacy land use constraints include a large estuary with several small-lot subdivisions serviced by unpaved roads, often perpendicular to the shore, including unpaved roads at Burnt Point. The core areas of Georgetown and Montague warrant specific policy attention as separate policy units (PUs).

Natural Assets and Heritage

Darnley Basin

includes the Brudenell River Provincial Park and the Montague Wildlife Management Area.

Parks and cultural facilities include Brudenell Courts (tennis), Brudenell Provincial Park, West Street Parks (playground), Georgetown tennis courts, the Georgetown elementary playground and gymnasium, AA MacDonald Memorial Park (playground), Brudenell River Golf Course, Spruce Street Park (playground), Montague Waterfront Park (playground), Clay Park (playground), the Montague consolidated field (soccer), Knox Dam (walking trail), Jean-Pierre Roma at Three Rivers

National Historic Site, the Georgetown Range Front Lighthouse and Rear Range Lighthouse, the St. Andrew's Point Pioneer Cemetery, St. Andrew's Point Park, and the Georgetown Waterfront Boardwalk.



Route 343 (Kayes Rd)



Georgetown



Georgetown wastewater treatment plant lagoon

Planning and Policy Framework

Three Rivers has municipal planning authority. Development on the coast is informed by the Environmental Protection Act, including a 15m buffer, with no additional building setbacks aside from the buffer. The Official Plan includes specific guidance permitting armouring, though such provisions are secondary to any provincial standards, and includes some policies regarding expansion of buildings within the sea level rise overlay.

There are no specific flood elevations or vertical setbacks established, but the Official Plan does include specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage.

Sediment transport

The estimated sediment transport averaged across the SMU is low. Sediment transport is highest and net westward along the headlands at the entrance of the two estuaries, including Burnt Point from which erosion feeds the beaches at Thrumcap Spit and Poxy Island off Georgetown.

Coastal Flood and Erosion Risk

While low plains and wetland shores are vulnerable to increasing flood hazards, infrastructure risk is low. Natural shoreline change is expected to occur through salt marsh migration along low plains and wetlands; following rising sea levels provided there is no coastal squeeze from infrastructure.

Historical shoreline change has been greatest at Poxy Island because it is an exposed sand spit, and it offers partial shelter to Georgetown waterfront and wastewater treatment plant lagoon.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline]

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	42 [1/km]	58 [1/km]	74 [1/km]	104 [1/km]	10 [<1/km]	53 [1/km]	140 [2/km]
Roads	1032 m [14 m/km]	1773 m [25 m/km]	2397 m [33 m/km]	4141 m [57 m/km]	642 m [9 m/km]	1807 m [25 m/km]	3421 m [48 m/km]

Proposed SMP Strategies

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
12.00 Darnley Basin	NAI	MR	MR
Rationale	Given the low risk levels, the initial strategy is No Active Intervention. Mid- to long-term Managed Realignment will reduce risk to the limited number of vulnerable assets, while preserving natural shorelines with increasing flood hazard.		
12.01 Georgetown waterfront -PU	R	R	R
Rationale	Resist strategy would be suitable for the waterfront, wharves and wastewater treatment plant as key infrastructure to the community, with flood mitigation measures as required.		
12.02 Lower Montague Rd -PU	R	MR	MR
Rationale	A Resist strategy is suitable for continued protection of the waterfront road and adjacent properties. In the mid- to long-term, increasing flood risks in the low-elevation sections will likely require gradual transition towards Managed Realignment strategies, notably along the eastern end. Future strategy for the Lower Montague wharf should be developed with the local authority.		
12.03 Montague waterfront -PU	R	R	R
Rationale	Resist strategy to maintain key infrastructure along the Town waterfront, with flood mitigation measures as required.		
12.04 Route 343 (Kayes Rd) -PU	R	R	R
Rationale	Resist strategy can be considered along the major road and associated seaward properties. The final strategy should be determined by the local community and the PEI Government depending on cost-benefits and local values.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.